Supplementary Notes I–IV

The lock, partition selection, the unimodular clock, and label-freeness — with round-by-round adjudications

July 2026 · The ETRG Program

Provenance disclosure: this document was constructed by four AI models (Claude Fable 5 — synthesis and implementation; DeepSeek-V4-Pro — derivations; GLM-5.2 — refereeing; Kimi-K2.7 — numerics) in an adversarial-review protocol, with a human originator (B. Kloosterman) supplying the thesis and direction. Every claim's attack history, including retractions by all four models, is preserved in the accompanying repository. Numerical results are reproducible from committed scripts. Treat accordingly.

ETRG-0 Addendum: The Lock Note

Sharpening S3: the factor of two as a null-sector readout, and A4 as a projection of A5. *Version 0.2 — July 2026. Self-contained; companion to ETRG-0_referee_packet.md. v0.1 → v0.2: revised after one adversarial round (DeepSeek-V4-Pro, GLM-5.2, Kimi-K2.7); adjudication appendix at the end. The central claims survived in weakened, more precise form; the "without Bianchi" novelty claim of v0.1 was wrong and is retracted.*

Why this note exists

Three independent first-pass constructions (July 2026), each responding blind to the constructive challenge in the referee packet, converged on the same architecture: an entropic-time temporal sector giving Newton, a spatial sector needed for the light-bending factor of two, and the requirement that the two sectors be locked at equal strength. All three *assumed* the lock. One (Kimi) stated the gap explicitly: "the factor of 2 is obtained because I assumed the temporal and spatial entropy gradients are equal — motivated but not proven."

This note upgrades the lock from assumption to argument — with the scope of each claim now stated exactly.

L-claims

L1 (The entropic input supplies exactly the trace-free sector). Every horizon-thermodynamic derivation of gravitational field equations feeds its entropic input through null surfaces: Jacobson's Clausius relation $\delta Q = T\,\delta S$ on local Rindler horizons (1995), the entanglement first law on causal diamonds whose null boundaries carry the modular flow (FGHMV 2014, Jacobson 2016), and Padmanabhan's null-surface variational program (2010, 2014). The reason is structural: entropy, temperature, and geometry meet operationally *only* at causal horizons (Bisognano–Wichmann, Unruh). The raw output of the entropic input is the null-projected equation

$$R_{ab}\, k^a k^b = 8\pi G\, T_{ab}\, k^a k^b \qquad \text{for all null } k^a \text{ at every point,}$$

a manifestly covariant condition (the null cone is frame-independent; both sides are scalars for each k). By the null-curvature lemma (Hawking & Ellis §4.3), this is *algebraically equivalent* to the trace-free part of the Einstein equations,

$$R_{ab} - \tfrac{1}{4}R\,g_{ab} \;=\; 8\pi G\left(T_{ab} - \tfrac{1}{4}T\,g_{ab}\right),$$

i.e. the unimodular form of gravity, in which the trace sector is undetermined and the cosmological constant enters only as an integration constant once the Bianchi identity and energy conservation are invoked (as in Jacobson 1995's own reconstruction; cf. the unimodular literature, Weinberg 1989 review, Ellis et al. 2011, and Padmanabhan's repeated emphasis that Λ arises as an integration constant in the emergent-gravity paradigm). **The precise claim is therefore not that the entropic input is "less than" the Einstein equations by a differential step, but that it is exactly the trace-free sector and is structurally silent on**

the trace/Λ sector. *Status: established literature — the round-2 prior-art sweep found this localization published: Alonso-Serrano & Liška, "Thermodynamics of spacetime and unimodular gravity," Int. J. Geom. Methods Mod. Phys. 19, 2230002 (2022), arXiv:2112.06301, show that three distinct thermodynamic derivations (Rindler wedges, causal diamonds, semiclassical sources) all yield unimodular/Weyl-transverse rather than fully diffeomorphism-invariant dynamics, with earlier observations cited therein. L1 is therefore imported, not contributed; the note's original content now lives entirely in L2–L4 (the observational readout) and L5 (the lock). This strengthens defensibility: the foundation is peer-reviewed.*

L2 (Light reads that sector directly). Static weak field, source rest frame, isotropic gauge:

$$ds^2 \;=\; -\,(1+2\Phi/c^2)\,c^2dt^2 + (1-2\Psi/c^2)\,\delta_{ij}\,dx^i dx^j.$$

In the static case Φ and Ψ coincide with the gauge-invariant Bardeen potentials, so nothing below depends on the gauge choice. Linearized: $R_{00} = \nabla^2\Phi/c^2$, $R_{0i} = 0$, $R_{ij} = \delta_{ij}\nabla^2\Psi/c^2 + \partial_i\partial_j(\Psi-\Phi)/c^2$. For a null direction $k = (1, \hat{n})$ (affine rescalings multiply both sides of L1's equation by λ^2 and cancel):

$$R_{ab}k^a k^b \;=\; \frac{1}{c^2}\left[\nabla^2(\Phi+\Psi) \;+\; n^i n^j\,\partial_i\partial_j(\Psi-\Phi)\right].$$

A light ray's total deflection is $\alpha = (1/c^2)\int\nabla_\perp(\Phi+\Psi)\,dl$ — the lensing observable depends on exactly the potential combination the null-null law sources. Light bending is the *direct* observational readout of the trace-free sector; the trace sector, the only part of the field equations the entropic input does not supply, never enters the lensing integrand. *Status: standard GR algebra; verified symbolically twice, by independent implementations (17/17 checks in* `lock_check.py`*, incl. the geodesic interpolation $a_\perp = -\nabla_\perp\Phi - (v^2/c^2)\nabla_\perp\Psi$ and $\alpha = 4GM/c^2b$).*

L3 (Corollary: in the static weak field, the trace-free sector suffices for γ = 1 and Newton jointly). Impose the L1 law for all null directions through every point, with a static source in its rest frame (the frame in which PPN γ is defined) and potentials decaying at infinity. For dust, $T_{ab} k^a k^b = \rho c^4$ is direction-independent, so decomposing the direction-dependent geometric term into trace and traceless parts forces the traceless Hessian of (Ψ−Φ) to vanish pointwise; the only decaying solution of $\partial_i\partial_j f = \tfrac{1}{3}\delta_{ij}\nabla^2 f$ is f = 0, hence **Ψ = Φ** (γ = 1), and the isotropic remainder gives $\nabla^2(\Phi+\Psi) = 8\pi G\rho$, hence $\boldsymbol{\nabla^2\Phi = 4\pi G\rho}$ — Newton's normalization. With anisotropic stress π_{ij} the same decomposition yields $\nabla^2(\Phi+\Psi) = 8\pi G(\rho+p)$ and $\partial_i\partial_j - \tfrac{1}{3}\delta_{ij}\nabla^2 = 8\pi G\pi_{ij}$ — *identical* to linearized GR, so the framework degrades exactly as GR does, neither better nor worse (adversarially confirmed, round 1).

Three honest deflations, from round 1: (i) given L1's equivalence lemma this corollary is elementary, not new mathematics; (ii) staticity, the rest frame, and asymptotic flatness are load-bearing hypotheses — the last is what excludes the Λ ambiguity here; (iii) the corollary adds no predictive content beyond linearized GR's trace-free sector. Its value is *locational*: it shows that everything light bending measures, including the Newtonian calibration it is compared against, lives inside the one sector the entropic input writes. *Status: proof checked symbolically and adversarially; survives as a corollary with stated hypotheses.*

L4 (The inversion — an explanation, not a prediction). With L1–L3 the traditional puzzle inverts. In any theory whose gravitational input is horizon thermodynamics, the primary object is the trace-free/null sector. Light — a null probe — reads that sector whole: $\alpha \propto \nabla_\perp(\Phi+\Psi)$. A slow body samples only the temporal projection, $a = -\nabla\Phi$, which for isotropic-stress sources carries exactly half of $\Phi+\Psi$. Nothing extra couples to fast particles; slow particles are the degenerate probe that hides half of the native structure. The measured 2:1 deflection ratio (1.75″ vs the 0.87″ of any pure clock-rate theory) is the sampling ratio of null versus timelike worldlines through one null-native law — the observational signature that the gravitational degrees of freedom being counted entropically live on null surfaces. This is offered as the *structural explanation* of the factor of two within the entropic program; it is emphatically not a new numerical prediction (the number is GR's), and any presentation that lets a reader believe otherwise fails packet checkpoint Q9. *Status: interpretation; checkable content is L1–L3.*

L5 (One generator, two faces — exact at entanglement equilibrium, approximate elsewhere, with a stated deviation scale). At a causal horizon with the field in (or infinitesimally near) the vacuum, the boost/modular generator $\hat{K}$ plays both roles at once:

- *Thermality face (temporal sector):* the reduced state is thermal in $\hat{K}$ (Bisognano–Wichmann) at the Unruh temperature; with Tolman–Ehrenfest, the position dependence of the modular temperature defines the lapse, $\sqrt{-g_{00}} = T_\infty/T(x)$ — axiom A4, derived rather than postulated *in this regime*.
- *Variational face (spatial sector):* $\delta S = \delta\langle\hat{K}\rangle$ with the single calibration $S = A/4G\hbar$ sources the spatial geometry (axiom A5).

One operator, one calibration constant, two projections: at entanglement equilibrium the sectors cannot be rescaled independently. **Scope restriction (round 1, scale corrected round 2):** away from equilibrium the two faces draw on *different* operators — the thermality face on the full modular Hamiltonian $\hat{K} = \hat{K}_{vac} + \Delta\hat{K}$ (nonlocal, non-geometric for excited states), the variational face on $\hat{K}_{vac}$ (through the vacuum-calibrated area law). The dimensionless measure of the drift is $\boldsymbol{\delta_{lock} = S(\rho\|\rho_{vac})/S_{BH}}$, bounded via Casini's form of the Bekenstein bound (Casini 2008) by $\boldsymbol{\delta_{lock} \le 8\pi\, \ell_P^2\, R\, \Delta E/(\hbar c A) = 2G\Delta E/(c^4 R)}$ for an excitation of energy ΔE and size R at a spherical horizon of that radius — i.e., *the excitation's own gravitational radius over the horizon radius* (round 1's $\Delta E\cdot\ell_P^2/A$ was not dimensionless and is superseded). For laboratory or astrophysical horizons this is astronomically small; for Planck-scale causal diamonds it is O(1). A round-2 refinement (DeepSeek), corrected in round 3 by lattice adjudication (`coherent_thermal_check.py`): coherent excitations preserve the reduced spectrum exactly ($\Delta S_{ent} = 0$; a unitary rotation of the wedge state), but at *matched energy* their relative entropy is **linear** in ΔE (quadratic in amplitude, and energy goes as amplitude squared), while spectrum-changing Gaussian excitations of the same energy have relative entropy suppressed to *second* order in ΔE — the first law's cancellation. The round-2 "quadratic suppression of coherent states" conflated amplitude order with energy order and is retracted. What the lock genuinely discriminates is which face responds: the spectral/entropy face is exactly blind to coherent drive while the modular-energy face is not — experimentally exploitable (packet P2, round-3 version). The lock is therefore an *equilibrium theorem with a quantified validity domain*, not an unconditional structural fact — this converts packet checkpoint Q7 from an open worry into a stated deviation scale, and marks where the framework's predictions could in principle depart from GR (strong non-equilibrium, Planck-scale diamonds). A further gap identified in round 1 and left open: the temporal sector's operational partition (Barontini's coarse-grained observed/unobserved split) and the spatial sector's partition (region/complement) are *different tensor factorizations*; the lock additionally requires that they coincide at horizons. Bisognano–Wichmann makes this plausible (the wedge algebra defines both) but it is not proven here — registered as new checkpoint **Q10** in the packet. *Status: theorem-grade at entanglement equilibrium; conjecture with stated deviation scale beyond it; Q10 open.*

L6 (What is imported — the honest ledger, revised). (a) The all-null-directions requirement is algebraically the trace-free Einstein equations (L1); the entropic derivation does not evade that content, it *is* that content — the retraction of v0.1's "without Bianchi" claim. What remains genuinely underived by the entropic input is the trace/Λ sector, recovered only via the Bianchi identity plus matter energy conservation, with Λ as an integration constant (the unimodular situation; partially defuses Q8's sting — Λ is unpredicted but unobstructed). (b) Asymptotic flatness excludes the Λ ambiguity in L3's regime. (c) The local-equilibrium, equation-of-state character of the Clausius input (Jacobson's own caveat) is inherited, now with L5's quantified deviation scale attached. (d) A Lorentzian causal structure — something for "null" to mean — is presupposed; A6's substrate question is untouched. (e) The coincidence of the two partitions (Q10) is assumed at horizons. The claim that survives all of this: *the specific metric sector that light bending measures is the sector the entropic law writes directly, and the 2:1 ratio is that law's structural signature.*

Requested attacks (round 2, when budget allows)

1. **Q10:** prove or refute that the Barontini-type coarse-grained partition and the region/complement entanglement partition define the same modular structure at a causal horizon (they must, for L5's two faces to share one $\hat{K}$).
2. **L5's deviation scale: done (round 2).** Round 1's $\Delta E \cdot \ell^2$_Planck/A failed the dimensional check; the corrected invariant is δ_lock = $S(\rho \| \rho$_vac$)/S$_BH $\leq 2G\Delta E/(c^4 R)$ (Casini bound), now in L5, with the coherent/thermal first-order discrimination as a bonus prediction.
3. **Prior art, second sweep: done (July 2026, web).** L1's trace-free localization *is* published (Alonso-Serrano & Liška 2022 — now cited; a win for defensibility). The L2–L4 readout inversion and the Q10 note's Takesaki selection mechanism were *not* found in either the model-knowledge sweep (round 1) or the web sweep (round 2); nearest kin are Padmanabhan's null-surface program, Eddington 1920, and Parzygnat–Russo's Bayesian-inversion/modular-group work. A third sweep by a human with journal access remains worthwhile before any submission.

Appendix: Round-1 adjudication (July 2026)

Attack	Source	Verdict	Disposition
"All null directions ≡ trace-free Einstein (Hawking–Ellis); 'without Bianchi' is vacuous"	GLM	**Sustained**	v0.1 novelty claim retracted; L1/L3/L6 rewritten around the trace-free localization, which is the defensible content
L3 proof gauge-dependent as written	GLM	Sustained (repairable)	Static Φ, Ψ are Bardeen potentials; gauge-invariance now stated in L2
L5 conjecture presented as claim	GLM	Sustained	L5 restricted to entanglement equilibrium; remainder labeled conjecture with deviation scale
Temporal and spatial sectors use *different factorizations*; coincidence at horizons unproven	GLM	Sustained, new	Registered as Q10; the sharpest new vulnerability found in round 1
"L3 frame-dependent, hence vacuous" (boosted-dust counterexample)	DeepSeek	**Overruled in part**	The null-cone quantification is frame-independent and the law covariant; the boost example varies the source while forgetting the metric's matching direction-dependence. Surviving residue: staticity/rest-frame/decay are load-bearing hypotheses, now stated in L3
"Anisotropic stress: degradation parametric with GR ⇒ no novel content"	DeepSeek	Sustained as calculation; verdict recontextualized	The identical degradation was a *requested consistency check* and it passed; the no-new-predictions point was already conceded (Q9, L6) — L4 reworded so it cannot be misread as a numerical prediction
Lock decouples off-equilibrium; mismatch ~ $\Delta E \cdot \ell^2\_\text{Planck}/A$	DeepSeek	**Sustained — most valuable finding of the round**	Q7 converted from open worry to quantified validity domain in L5

Attack	Source	Verdict	Disposition
17/17 symbolic verification of L2/L3 algebra + geodesic interpolation + $4GM/c^2b$	Kimi	Confirmed	`lock_check.py`, committed c2883cb; independent of the local check `ricci_check.py`

References to add to the packet

A. Alonso-Serrano & M. Liška, Int. J. Geom. Methods Mod. Phys. 19, 2230002 (2022), arXiv:2112.06301 (**the published form of L1**: thermodynamics of spacetime yields unimodular gravity) · T. Padmanabhan, Rep. Prog. Phys. 73, 046901 (2010) · T. Padmanabhan, Gen. Rel. Grav. 46, 1673 (2014) (null-surface variational principle) · Hawking & Ellis, *The Large Scale Structure of Space-Time*, §4.3 (null-curvature lemma) · S. Weinberg, Rev. Mod. Phys. 61, 1 (1989) and G. F. R. Ellis et al., Class. Quantum Grav. 28, 225007 (2011) (unimodular gravity; Λ as integration constant) · A. S. Eddington, *Space, Time and Gravitation* (1920) (the classic space+time-curvature account of the factor of two). Any prior statement of the L4 inversion found by reviewers supersedes this note's framing claims.

ETRG-0: The Q10 Note — Two Partitions, One Generator

Claim: causal confinement plus operational consistency *selects* the factorization, and at a horizon every admissible coarse-graining inherits the boost generator. Q10 is answerable, and the answer also blunts Q3. *Version 0.2 — July 2026. Authored by Claude (Fable 5, Anthropic) as an equal-contributor pass; submitted for the same adversarial review as everything else in this program. v0.1 → v0.2: N2's consistency premise repaired after GLM's round-2 circularity attack (adjudication appendix at the end); N5's cosmological limitation stated; lattice verification results added.*

The problem (from round 1)

Checkpoint Q10 (GLM's find): the temporal sector's operational partition — Barontini's coarse-grained observed/unobserved split, which supplies entropic time — and the spatial sector's partition — region/complement across a surface, which supplies the entanglement first law — are *different tensor factorizations* with, in general, different modular generators. The lock note's L5 needs them to define the same modular structure at causal horizons. Bisognano–Wichmann makes that plausible; nothing proves it. Separately, Q3 (Albrecht–Iglesias clock ambiguity) asks what forbids re-partitioning the universe to game entropic time.

This note argues both questions have the same answer, and that it is a theorem-shaped answer, not a plausibility.

N-claims

N1 (Observers at horizons are subalgebras of the wedge). An observer confined to the exterior of a causal horizon can operate only on the wedge algebra $A(W)$: causality is not a modeling choice but a physical restriction on which operations exist for them. Any coarse-graining such an observer implements — finite resolution, mode binning, restricted instrument set — is a subalgebra $N \subseteq A(W)$ together with an assignment of reduced statistics. The “bright/dark” split of a Barontini-class construction, transplanted to a horizon, is therefore not an arbitrary tensor factorization of the global Hilbert space; it is constrained to live inside the geometric partition from the start. *Status: physical framing; the content is that Q3’s “arbitrary re-partition” is not available to any causally situated observer.*

N2 (Operational consistency forces modular covariance — Takesaki). For the observer’s coarse-grained bookkeeping to be *self-consistent* — meaning there exists a conditional expectation $E: A(W) \to N$ that preserves the state ω (so that expectation values of coarse observables can be computed within the coarse description without knowing which fine details were discarded, and repeated coarse measurements don’t systematically drift from the fine statistics) — Takesaki’s theorem (J. Funct. Anal. 9, 306 (1972)) applies: **a normal state-preserving conditional expectation onto N exists if and only if N is invariant under the modular flow σ_t^ω of $(A(W), \omega)$.** Moreover, when E exists, the modular flow of the restricted state on N *is* the restriction of the modular flow of $A(W)$: $\sigma_t^{\omega|N} = \sigma_t^\omega|_N$. For the vacuum on a Rindler wedge, σ_t^ω is the boost (Bisognano–Wichmann). Therefore every operationally consistent coarse-graining at a horizon has, as the modular generator of its own reduced state, *the restriction of the same boost generator $\hat{K}$.* One generator, by theorem — not by the assumption round 1 correctly flagged. *Status after round 2: the v0.1 premise was correctly convicted of circularity (GLM): “consistent bookkeeping ⇒ state-preserving conditional expectation” defined consistency as the very condition Takesaki converts to modular invariance; an observer dephasing in a non-commuting basis keeps drift-free records of everything they track, refuting the premise as stated.* ***Repaired premise (v0.2, dynamical):*** *the observer’s coarse-graining must commute with their own time evolution — coarsen-then-evolve = evolve-then-coarsen — or coarse predictions drift against coarse records as time passes; this demand is operational and modular-theory-free. The physics input is then that a horizon observer’s evolution is generated by the boost, which IS the modular generator (Bisognano–Wichmann/Unruh). Commuting with $e^{i\hat{K}t}$ for all t forces the coarse-graining onto $\hat{K}$-invariant structure — same conclusion, no circularity. The residual assumption to attack: that “no drift under your own dynamics” is the right minimal notion of a functioning clock-keeping observer.*

N3 (First-order lemma: the coarse first law equals the fine first law for modular-covariant coarse-grainings). Let $\rho = e^{-K}/Z$ be the reduced vacuum on a region, and $\rho(\varepsilon) = \rho + \varepsilon\, \delta\rho$ a perturbed family. The fine first law is $dS/d\varepsilon|_0 = \mathrm{Tr}(\delta\rho\, K) = d\langle K\rangle/d\varepsilon|_0$ (E7). Now let D be dephasing in the eigenbasis of K (or in binned modular-energy shells): $D[X] = \Sigma_s P_s X P_s$ with $[P_s, \rho] = 0$. Define the coarse entropy $S_{cg}(\varepsilon) = S(D[\rho(\varepsilon)])$. Then:

$$\frac{dS_{cg}}{d\varepsilon}\Big|_0 = -\mathrm{Tr}\big(D[\delta\rho]\ln\rho\big) = -\mathrm{Tr}\big(\delta\rho\, D[\ln\rho]\big) = -\mathrm{Tr}(\delta\rho \ln \rho) = \frac{d\langle K\rangle}{d\varepsilon}\Big|_0 ,$$

using $D[\rho] = \rho$, self-adjointness of D, and $[\ln \rho, P_s] = 0$. **The coarse-grained entropy obeys the same first law, with the same modular Hamiltonian and the same calibration, to first order.** For a coarse-graining basis B that does not commute with K, $D\_B[\rho] \neq \rho$ and $dS\_cg/d\varepsilon = -\mathrm{Tr}(\delta\rho\, D\_B[\ln D\_B[\rho]]) \neq d\langle K\rangle/d\varepsilon$ generically: the lock breaks. So the lock between the temporal face (coarse entropy bookkeeping) and the spatial face (fine first law) holds *precisely for* the coarse-grainings N2 says an admissible observer must use. Binning into shells of finite width ΔK adds corrections controlled by ΔK; they vanish as the resolution refines. *Status: elementary proof, stated here for the record; lattice verification requested (prediction below).*

N4 (The two faces are one bookkeeping — closing the circle with Clausius). Under N2+N3, an admissible horizon observer's entropic-time increments track coarse entropy exchange, $d\tau \propto |dS\_cg|$; N3 says $dS\_cg = d\langle\hat{K}\rangle$ — modular energy flux; and modular energy flux through a horizon at Unruh temperature is exactly the Clausius input $\delta Q = T\,\delta S$ of the spatial sector's derivation (Jacobson). The temporal face's clock and the spatial face's source are not two quantities that happen to agree; they are *one ledger read twice*. This is the precise content S1 ("one currency") was reaching for, now with the mechanism named: modular covariance of admissible coarse-grainings. *Status: interpretation built on N2+N3.*

N5 (What this does to Q3). The clock-ambiguity objection assumed all tensor factorizations are equally available. N1 removes the non-causal ones for any situated observer; N2 removes the causally available but operationally inconsistent ones (no state-preserving conditional expectation → the observer's coarse records cannot be maintained without systematic drift, i.e. that partition does not support a functioning clock at all — a resonance with A3, clocks cost consistency, not just entropy). What survives is a class of admissible partitions all sharing one modular generator at the horizon. Predictions are therefore partition-independent *within the admissible class* — which is the strongest form of factorization-independence this program can claim, and, we submit, the right form: Q3's "gameable time" partitions are exactly the ones in which no observer can keep books. *Status after round 2: sustained only for horizon-type observers. GLM's attack stands for the cosmological case Q3 actually targets: a comoving FRW observer has no wedge, and in de Sitter the comoving-mode factorization is not anchored to the static-patch algebra. The generalization target is the causal **diamond** of any finite-lifetime observer — every physical observer has one, FRW is conformally flat, and conformal diamonds have geometric modular flow (Hislop–Longo), so the conformal sector is tractable; the non-conformal general case folds into Q2/Q7/Q8 and stays open. Q3 is blunted for horizon/diamond observers, not defeated in general.*

Lattice prediction (testable now, free fermions)

On a free-fermion chain in its ground state with interval A, modular modes from diagonalizing the interval correlation matrix $C\_A = U\, \mathrm{diag}(n\_k)\, U^\dagger$, and a weak global perturbation of strength ε:

1. **Fine:** $dS\_A/d\varepsilon = d\langle K\_A\rangle/d\varepsilon$ (already verified in Demo 3 / round 1).
2. **Coarse-modular:** define $\tilde{n}\_k(\varepsilon) = [U^\dagger C\_A(\varepsilon)U]\_{kk}$ (dephasing in the modular mode basis) and $S\_mod(\varepsilon) = \Sigma_k\, h(\tilde{n}_k)$ with h the binary entropy. Prediction: $dS\_mod/d\varepsilon = dS\_A/d\varepsilon$ exactly at first order (equal slopes over ≥2 decades of ε; deviation slope 2, i.e. $O(\varepsilon^2)$).
3. **Coarse-site (control):** $S\_site(\varepsilon) = \Sigma_i\, h([C\_A(\varepsilon)]\_{ii})$ (dephasing in the site basis, which does not commute with K_A). Prediction: $dS\_site/d\varepsilon \neq dS\_A/d\varepsilon$ by an O(1) ratio, not converging with ε.

A pass on (2) with a fail on (3) is the lattice avatar of N2/N3: the lock survives coarse-graining if and only if the coarse-graining is modular-covariant.

Status: verified (round 2, `q10_lattice_check.py`). At filling 0.4, $L = 200$, $\ell = 40$: first law pairing/ $\delta S\_fine \to 0.99991$; coarse-modular ratio $\to$ 0.99992 (predicted 1); coarse-site control ratio $\to$ 4.216 (predicted $\neq 1$); residual slope 2.0000 (predicted 2). Two honest annotations. (i) The first run, at half filling per the original spec, gave a *null result*: a pure potential perturbation is particle-hole-odd there, so the entire first-order response vanishes by symmetry ($S(\varepsilon) = S(-\varepsilon)$) and the run probed only the second-order regime — the spec's error, caught and fixed by moving off half filling. (ii) The sweep shows the modular lock degrading for $\varepsilon \gtrsim 10^{-3}$ exactly as second-order terms overtake the linear response — the lattice image of L5's "equilibrium theorem with a finite validity domain," and a reminder that N3 is a first-order statement only.

Requested attacks

1. **N2's operational premise.** "Consistent bookkeeping ⇒ state-preserving conditional expectation exists" is the weakest joint. Construct an observer whose coarse-graining is physically reasonable (implementable, repeatable, drift-free in the quantities they track) yet admits no state-preserving conditional expectation — that would break the selection principle. Note the premise requires preservation of the *actual* state ω, not all states; argue whether that is the right operational demand or too strong/too weak.
2. **Type III honesty.** Wedge algebras are type III_1: no density matrices, no P_s projectors onto "eigenspaces of K". N3's proof is type I (lattice-regularized). Does the N2 selection argument survive in the continuum (Takesaki's theorem itself is fine in type III; the dephasing lemma needs restating via relative entropy or Connes cocycles)?
3. **Barontini fidelity.** Is the actual Birmingham bright/dark split (mode-space, not region-space) modular-covariant for the state they prepare? If it is not even approximately so, N2 predicts their entropic time should *drift* from any modular/thermal time — is that drift visible in their data, and does its absence/presence test this note?
4. **Beyond horizons.** N2 hands us the generator only where modular flow is geometric. State the sharpest version of what remains unlocked for generic regions (this is Q7's territory; be precise about what N2 does *not* buy).
5. **σ corollary (optional).** Modular flow is dimensionless (KMS at inverse "temperature" 2π); physical proper time enters via ħ and the local temperature. Does N2+N4 fix the packet's calibration constant σ = σ(ħ, k_B, T) up to a pure number? If yes, A2 becomes parameter-free; derive or refute.

Relation to existing claims

Strengthens: L5 (supplies the missing "same factorization" argument at horizons — Q10), S1 (one currency, now with mechanism), A3 (clocks need consistency), Q3 defense (admissibility selection). Leaves untouched: Q1 (constraint algebra), Q2 (out-of-equilibrium lapse), Q6, Q8, Q9's general worry. New exposure: if attack 3 finds the Barontini split badly non-modular yet their entropic clock works fine anyway, N2's operational premise is empirically undermined — this note, unlike most theory notes, can lose to existing data.

Appendix: Round-2 adjudication (July 2026)

Attack	Source	Verdict	Disposition
N2 circular: "consistency" defined as Takesaki's own condition; non-commuting dephasing keeps drift-free records with no state-preserving conditional expectation	GLM	**Sustained**	Premise repaired to the dynamical form (coarse-graining must commute with the observer's own evolution); physics input relocated to BW/Unruh; residual assumption named
Q3 defense horizon-bound: fails for comoving FRW observers (no wedge) and de Sitter comoving modes (wrong algebra)	GLM	Sustained in part	N5 rescoped: blunted for horizon/diamond observers only; causal-diamond generalization named as the research direction (conformal sector tractable via Hislop–Longo); general case folds into Q2/Q7/Q8
Lattice test of N3	Kimi (script) + local run	**Verified** (after fixing a particle-hole symmetry null in the original half-filling spec)	First law 0.99991; modular ratio 0.99992; site control 4.216; residual slope 2.0000
Deviation scale for the L5 lock (dimensional repair of round-1's $\Delta E \cdot \ell^2_P/A$)	DeepSeek	Delivered	Correct invariant $\delta_{lock} = S(\rho \| \rho_{vac})/S_{BH} \leq 2G\Delta E/(c^4 R)$ via the Casini bound — the excitation's gravitational radius over the horizon radius; folded into the lock note L5
Coherent-state evasion: relative entropy quadratic in displacement; coherent excitations preserve the lock at first order, thermal excitations break it linearly	DeepSeek	Delivered	Upgraded into the packet's P2 as a coherent-vs-thermal experimental discriminator

References

Takesaki, J. Funct. Anal. 9, 306 (1972) · Accardi & Cecchini, J. Funct. Anal. 45, 245 (1982) (generalized conditional expectations, for attack 1) · Parzygnat & Russo, "Bayesian inversion and the Tomita–Takesaki modular group" (2021) (state-preserving conditional expectations as Bayesian inverses — nearest adjacent lineage found for N2; no gravitational application located) · Bisognano & Wichmann (1976) · Connes & Rovelli (1994) · Šafránek, Deutsch, Aguirre, PRA 99, 010101 (2019) (observational entropy, for N3's coarse entropy) · Jacobson, PRL 75, 1260 (1995) · Barontini et al., arXiv:2509.07745.

ETRG-0: The Unimodular Clock Note — Q1, and the Time that Comes with the Trace-Free Sector

Claim (v0.2, post-review): the constraint-algebra worry (Q1) localizes to off-equilibrium corrections with a stated covariance criterion; the unimodular formulation the entropic input *permits* carries a global clock — an available structure, not an automatic consequence. *Version 0.2 — July 2026. Authored by Claude (Fable 5, Anthropic), round 3; revised after adversarial review by DeepSeek-V4-Pro and GLM-5.2 (adjudication appendix below). v0.1's HKT-uniqueness armor is retracted; U5's two-clocks conjecture is withdrawn as ill-posed.*

U-claims

U1 (Q1 at equilibrium: closure holds for the algebra the theory actually has). At entanglement equilibrium the derived dynamics is the trace-free Einstein sector with Λ as an integration constant — unimodular gravity. Its Hamiltonian structure is known and closes (Henneaux & Teitelboim 1989): the gauge group is reduced (volume-preserving spatial diffeomorphisms; spatially constant lapse), and the zero mode of the Hamiltonian constraint is promoted from constraint to true Hamiltonian. **Retraction (round 3):** v0.1 additionally invoked Hojman–Kuchař–Teitelboim uniqueness as armor. That was a category error — HKT is a theorem about representations of the *full* hypersurface-deformation algebra, which unimodular gravity restricts away; one cannot claim the benefits of an algebra the theory does not carry (DeepSeek, finding 2). The honest statement: closure is established directly for the restricted algebra, the restriction is a *feature the entropic derivation must own* (it is exactly the trace-silence of L1), and the quantum inequivalence of unimodular vs full GR (trace-mode fate, measure, Λ superselection) is an open exposure, not a technicality. *Status: established results, correctly scoped.*

U2 (Two layers: covariant derivation, state-selected clock). The Q1 threat "a preferred foliation deforms the algebra" conflates two different structures, and keeping them apart is the defense:

- *Derivation layer.* The entropic input is imposed on **all** local causal horizons/diamonds — every point, every boost frame (Jacobson's construction; the same all-null-directions quantification that powered the lock note's L3). No foliation is chosen; that quantification over all frames is precisely why the output is a tensor equation. A derivation that quantifies over every frame cannot smuggle in one frame.

- *Description layer.* The operational entropic clock (A2/A4: Barontini τ, Tolman lapse) is supplied by the *state* of a given equilibrium solution, as the CMB supplies a cosmological rest frame — solution-level structure, absent from the field equations.

DeepSeek's round-3 attack (a self-referential ξ^μ-projected field equation) targets a construction ETRG-0 does not make: the projection onto a single state-selected foliation never appears at the derivation layer. **What survives of the attack:** A7's state-dependent lapse $\tilde{N}[\psi]$ must remain purely descriptive when matter back-reacts on geometry. If $\tilde{N}[\psi]$ ever enters the *source terms* of the geometry sector, the two layers collapse into DeepSeek's self-referential loop and the algebra deforms. Packet Q5's "pure reparameterization" defense must therefore be proven in the back-reacting case, not just for fixed backgrounds — registered as the sharpest surviving Q1 exposure. *Status: two-layer distinction is sound; the back-reaction proof is open and named.*

U3 (Where Q1 can still bite: off-equilibrium corrections, with the obstruction now explicit). Round 2 gave the lock's deviation scale $\delta_{lock} = S(\rho \| \rho_{vac})/S_{BH}$. If effective field equations acquire corrections at this order, Q1 becomes: are they covariant? The assignment (causal diamond → relative entropy) is covariant, but DeepSeek's round-3 analysis (finding 3) shows the step from diamond-wise scalars to a tensorial $\delta G_{\mu\nu}$ is where covariance lives or dies, with three named failure modes: (i) the measure on the space of diamonds through a point must be metric-covariant, not foliation-weighted; (ii) the small-diamond expansion's *shape/orientation derivatives* introduce a directional structure that can smuggle the foliation back in; (iii) the correction must conspire to stay trace-free, or it conflicts with the unimodular structure in which the trace is an integration constant — a nontrivial consistency condition. Sharpened Q1 (round 4+): derive the leading δ_{lock} correction from the FGHMV setup and check items (i)–(iii). *Status after round 4 (DeepSeek derivation, adjudicated): the computation exists in the literature — FHHPRV 2017 (arXiv:1705.03026) show the second-order relative entropy produces the quadratic completion of the source, a local covariant bilinear functional. Verdicts: (ii) no foliation dependence at leading order (angular integration over diamond orientations cancels directional terms); (iii) the second-order source has a trace piece, but it enters via Bianchi + conservation with Λ still an undetermined integration constant — the unimodular character persists at all orders (DeepSeek's contrary "unimodular is linear-order-only" framing was overruled by its own equations); (i) the measure over diamond space is fixed by conformal symmetry in the CFT setting and remains the honest open obstruction for generic spacetimes. Net: Q1's off-equilibrium threat is resolved at leading order in the conformal case; the generic-measure question is what remains.*

U4 (The unimodular clock — available, not automatic). In the Henneaux–Teitelboim formulation, Λ is a conserved quantity whose conjugate is accumulated four-volume — "cosmological time" T — and quantum-mechanically the theory evolves by a Schrödinger equation in T rather than being frozen (Unruh 1989; Unruh & Wald 1989; in quantized form, Smolin 2009/2011, arXiv:0904.4841, 1008.1759). **Correction (round 3, GLM objection 3):** the entropic derivation fixes the *field equations* (trace-free, Λ undetermined); it does not fix the *Hamiltonian formulation*. Alonso-Serrano & Liška's preferred reading is Weyl-transverse gravity, which does not carry the HT clock. So the clock is a formulation the trace-silence makes *available*, not a consequence it *generates* — v0.1's "the silence generates the clock" overstated. The interesting residual question is whether anything on the operational side selects the HT formulation: A2/A3 demand that observers possess clocks; HT-unimodular is the formulation in which the gravitational trace sector *is* one.

That selection argument is a research direction, not a result. **Kuchař's critique imported in full (GLM finding 3):** the unimodular clock is a *trade*, not a partial win — one global time (not many-fingered), non-local (four-volume requires the whole spacetime), a different observable structure, and a semiclassical recovery problem; choosing this formulation forecloses standard routes to the remaining problems of time. *Status: established components; the thermodynamic→unimodular→clock bridge appears unpublished (GLM: InspireHEP searches return zero; components live in disjoint literatures), but it is a bridge between formulations, not a derivation.*

U5 — withdrawn. v0.1 conjectured that entropic time τ and unimodular time T are monotonically related, with a Bose–Hubbard toy test. Both reviewers independently found this ill-posed: the two clocks live in different frameworks (a state-dependent quantum-information flow parameter vs a classical phase-space coordinate), and a Bose–Hubbard simulator has no intrinsic four-volume — any analogue is arbitrary, so the test "cannot fail for the right reason" (GLM objections 1–2; DeepSeek concurring). The conjecture is withdrawn as a claim. What remains is a *question*: whether a semiclassical bridge regime exists in which both are simultaneously definable. Constructing such a regime is a prerequisite to even stating the conjecture, and no candidate is currently known. *Status: withdrawn; prerequisite named.*

Requested attacks (round 4, when opened)

1. **U2's surviving exposure:** prove or refute that A7's Ñ[ψ] remains outside the geometry sector's source terms under semiclassical back-reaction (this is packet Q5 upgraded to the gravitating case). A proof that it cannot stay outside kills the two-layer defense and with it the program's Q1 answer.
2. **U3's computation:** the leading δ_lock correction from the FGHMV setup, checked against DeepSeek's three failure modes — especially the trace-free consistency condition (iii), which is a sharp yes/no.
3. **U4's selection question — answered (round 4, GLM finding 5):** A2/A3 do *not* select Henneaux–Teitelboim: both formulations satisfy the operational clock requirement (τ is available in either); the choice is conventional at the classical level. The formulations differ quantum-mechanically — HT has a canonical gravitational time (Schrödinger evolution of the gravitational sector in T), WTG stays frozen at the gravitational level — but ETRG-0's dynamics uses τ, not T, so the difference is not currently operationalized. U4 accordingly stays a research direction, with the one stronger question that could revive it: whether self-consistency of the A2→A4→A5→A7 loop requires an unfrozen gravitational sector (no such argument exists; it would need to show the WDW frozen formalism is inconsistent with a state-dependent matter clock).

Appendix: Round-3 adjudication (July 2026)

Attack	Source	Verdict	Disposition
U1's HKT citation is a category error (theorem about the full deformation algebra; unimodular restricts it)	DeepSeek	**Sustained**	HKT armor retracted; closure claimed only for the restricted algebra (Henneaux–Teitelboim); quantum inequivalence named as exposure
U2's gauge argument is "wishful": self-referential foliation lock-in under back-reaction	DeepSeek	**Overruled in main thrust, sustained in residue**	The derivation layer quantifies over all frames (no foliation to leak); but the A7 back-reaction proof is a genuine open exposure, now named as round-4 attack 1
U3's criterion necessary but not sufficient; three explicit obstruction modes	DeepSeek	Sustained (constructive)	All three folded into U3; trace-free consistency (iii) flagged as the sharp test
U4's clock is a formulation choice, not a consequence (WTG carries no HT clock)	GLM	**Sustained**	U4 reworded from "generates" to "makes available"; the operational-selection question named as the residual research direction
Kuchař 1991 kills more of U4 than admitted (trade, not partial success)	GLM	Sustained	Full import: global-only, non-local, different observables, foreclosed standard routes
U5 ill-posed (category error; toy test vacuous)	GLM + DeepSeek	**Sustained**	U5 withdrawn as a claim; prerequisite (a bridge regime) named
U4 prior art: bridge unpublished; components disjoint (Alonso-Serrano & Liška ↔ Smolin/Unruh-Wald never connected; InspireHEP cross-searches empty)	GLM	Delivered	Novelty claim retained for the bridge only, with its deflated status (a bridge between formulations)

Attack	Source	Verdict	Disposition
Round-2 coherent/thermal claim: at matched energy, hierarchy reverses (coherent S_rel linear in ΔE, Gaussian-noise S_rel quadratic); discriminator is which leg moves	Fable (lattice, `coherent_thermal_check.py`, correcting DeepSeek round 2 and Kimi's draft formula)	**Verified numerically**	Packet P2 and lock-note L5 corrected; round-2 "quadratic suppression" retracted

References

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ETRG-0: The Label-Freeness Note — A7 Under Back-Reaction

Claim: the entropic-time lapse cannot deform the constraint algebra because no equation of the theory consumes the time label — with one named discipline the formulation must maintain, and one toy experiment that shows exactly where the leak would occur. *Version 0.2 — July 2026. Authored by Claude (Fable 5, Anthropic), round 4. Answers the surviving Q1 exposure named in ETRG-0_unimodular_clock_note.md (round-4 attack 1); upgrades packet Q5 to the gravitating case. v0.1 → v0.2 after GLM review and the toy runs: R1 conditioned to equilibrium (the excited-state case merges with U3's burden), the R3 audit corrected on A4, the rule refined (state-computed rates are safe; only label-derivatives leak), and the toy extended with GLM's lapse-coupling control — which my refined rule survived, after a stiffness scare (adjudication appendix).*

R-claims

R1 (The geometric input is label-free). The entanglement first law $\delta S_A = \delta\langle K_A\rangle$, the entanglement-equilibrium condition, and their Einstein-equation consequences are functionals of *states on causal diamonds*. Nothing in them references how any observer parameterizes evolution. Semiclassical sourcing likewise: $\langle T_{\mu\nu}\rangle$ is a state functional. Geometry, in this construction, is determined by the matter state trajectory as an *unparameterized curve* in state space. *Status after round 4 (GLM finding 1, sustained in part): exact at equilibrium — for the vacuum, Bisognano–Wichmann makes the diamond restriction independent of the Cauchy slicing. For excited states the reduced state does depend on the slicing used to factorize, so R1's label-freeness there is exactly as strong as U3's covariance result and no stronger: R1 and*

U3 stand or fall at the same joint. DeepSeek's round-4 derivation covers that joint at leading order in the CFT setting (the S_rel correction is a local covariant bilinear — FHHPRV 2017); the generic-spacetime case remains open with U3's obstruction (i).

R2 (A7 is a relabeling of the same curve). A7's equation $i\hbar\, d|\psi\rangle/d\tau = \tilde{N}[\psi]\, \hat{H}|\psi\rangle$ with $\tilde{N} = |dS/dt|^{-1}$ is, by construction, the ordinary Schrödinger evolution with $d\tau = |dS/dt|\, dt$: the same state-space orbit traversed with a different speed function. (Equivalently: $dS/d\tau = \pm 1$ identically — entropic time is arclength in entropy.) Since R1's machinery consumes only the orbit, $\tilde{N}[\psi]$ cannot enter the geometry. DeepSeek's round-3 self-referential loop (state → modular flow → τ → foliation → geometry → state) therefore cannot close through this channel: geometry responds to the state, the label responds to the state, but geometry never responds to the label. The loop's dangerous arrow (τ → geometry) does not exist in the theory. *Status: the core claim; its proof is the conjunction of R1 (no consumer of the label) and the definition of A7 (label-only modification).*

R3 (The one place a leak could occur — the modular-normalization rule). A leak requires an equation that consumes a *rate with respect to operational time*. Audit of where rates enter: Jacobson's Clausius input uses heat flux and temperature both defined per unit *boost/modular* parameter of the local horizon — a geometric structure of the diamond, not anyone's bookkeeping — so it is internally normalized and label-free. The audit of A2–A7: A2's τ consumes $dS/d\varphi$ (internal, relational, feeds no geometric equation); A4's lapse is a temperature *ratio* (Tolman), label-free; A5 consumes state variations; A7 consumes the label only. The discipline the formulation must maintain forever after: **modular quantities may only be normalized by modular flow, never by entropic time.** A future variant that, e.g., defined local temperature per entropic-time tick would leak the label into the dynamics and deform the algebra — this is a design constraint, stated here so violations are checkable by inspection. *Status after round 4: the audit as first written was too quick on A4 (GLM finding 2, sustained in part). A4's lapse is $T_\infty/T(x)$, and T_∞ is the temperature referred to the global entropic clock — the ratio is invariant under global clock rescaling, but calling it "label-free" requires the identification of thermal time with entropic time, which is exactly L5's thermality face: **exact at equilibrium/horizons (where Bisognano–Wichmann fixes the modular temperature independently of τ), open beyond (Q2)**. A4's compliance with the rule is therefore conditional, not unconditional, and the calibration should always be anchored at a horizon. Separately, the toy's run-5 result refines the rule itself: **rates computed from the state via the Hamiltonian (including the lapse $\tilde{N}[\psi] = |dS/dt|^{-1}$ itself) are label-free and may enter couplings; only derivatives taken with respect to the label in use leak.** dS/dt is a state functional; $dS/d\lambda|_{\lambda=\tau} \equiv \pm 1$ is not the same object, and that difference is the entire leak.*

R4 (Physical clocks gravitate; labels do not). A3 says clocks dissipate: any physical clock has stress-energy, and that stress-energy sources geometry covariantly like all matter. Back-reaction of the bookkeeping *apparatus* is ordinary, covariant physics. The distinction doing the work: the clock's **matter** gravitates; the clock's **label** does not. Conflating these is what makes "the observer's time affects the geometry" sound threatening; separating them dissolves the threat. *Status: conceptual, but load-bearing for Q1 and Q5.*

R5 (Consequences). (i) Q1: with R1–R4, the round-3 two-layer defense upgrades from argument to proof-sketch — the equilibrium constraint algebra cannot be deformed by A7 because the theory contains no consumer of the A7 label. (ii) Q5 (signaling): nonlinearity confined to the label cannot transmit information, because no physical coupling reads the label — the packet's "pure reparameterization" defense now extends to the gravitating case, conditional on R3's rule holding. (iii) The remaining Q1 threat is exactly and only U3's off-equilibrium δ_{lock} corrections (round-4 attack 2, running in parallel). *Status: follows if R1–R3 survive review.*

The toy experiment (specification)

A solvable demonstration of precisely where the leak lives. Two-sector Bose–Hubbard mini-universe (Demo-1 style, smaller: $N \approx 60$), time-independent $\hat{H}$. Four runs:

1. **Baseline:** evolve $\psi(t)$ exactly; compute bright-sector $S(t)$, $\tau(t) = \int|dS|$.
2. **A7 reconstruction:** integrate $i\hbar\, d\psi/d\tau = \tilde{N}[\psi]\hat{H}\psi$ from internal data; verify the orbit coincides with run 1 (fidelity at matched states ≈ 1). [Replicates Demo 1.]
3. **State-functional feedback (the safe channel):** add $\hat{H} \to \hat{H} + g\cdot f(S_{bright}[\psi])\cdot\hat{V}$ — a crude "geometry responds to the state." Evolve in t; reconstruct in τ with the same state-functional coupling. Prediction: orbits still coincide to integrator precision — back-reaction through the state does not break reparameterization invariance.
4. **Label-consuming feedback (the leak, as control):** couple instead $g\cdot(dS/d\lambda)\cdot\hat{V}$ where λ is *the evolution parameter actually used* (dS/dt in the t-run, $dS/d\tau \equiv \pm1$ in the τ-run). Prediction: orbits diverge, and the divergence grows with g — the lattice image of a theory that violates R3's rule.

A pass on 3 with a fail on 4 is the sharpest possible toy statement of R2/R3: the state channel is safe, the label channel leaks, and ETRG-0's axioms use only the state channel.

Results (round 4, `label_freeness_toy.py`, run locally after Kimi's session died). Run 2 (A7 reconstruction): mean fidelity 0.99985. Run 3 (state-functional feedback): 0.99983 — indistinguishable from baseline; the state channel is safe under back-reaction. Run 4 (label-consuming control): mean 0.898, min 0.228 — and crucially **step-size independent** (0.898 → 0.894 under step halving): a genuine dynamical difference, the leak made visible. Run 5 (GLM's proposed sharper control — couple the lapse $\tilde{N}[\psi]$ itself): at moderate coupling (lapse cap 2) fidelity 0.99984, fully safe as the refined rule predicts, since $\tilde{N}[\psi]$ is a state functional and both parameterizations then integrate the *same* autonomous state-space ODE. At strong coupling (cap 20) the runs diverge numerically (mean 0.805), but this divergence *shrinks under step halving* (0.805 → 0.871) and vanishes at moderate coupling — stiffness, not physics, concentrated at stasis points where the entropic-time coordinate is singular. The toy thus also reproduces A7's own reading: the τ-coordinate description fails numerically exactly where it is a coordinate singularity. Min-fidelity dips (~0.95) in the safe runs are the known stasis-regularization error of Demo 1, halving with step size as integrator error should.

Requested attacks

1. Find a label hidden in R1's machinery — e.g., does the *choice of Cauchy slicing* used to define "the state" reintroduce a foliation at the semiclassical level, and if so, does covariance of $\langle T_{\mu\nu}\rangle$ under slicing changes fully absorb it?

2. Break R3’s audit: identify an existing ETRG-0 joint that already normalizes a modular quantity by entropic time.
3. R4’s separation (matter vs label) assumes the clock’s dissipation rate does not itself enter any geometric equation as a rate-per-label. Construct a scenario where clock thermodynamics (A3) forces a label-consuming coupling.
4. The toy: is run 4 a fair image of the feared leak, or a strawman? Propose a sharper control if so.

Appendix: Round-4 adjudication (July 2026)

Attack	Source	Verdict	Disposition
R1 hides a label: excited-state restrictions depend on Cauchy slicing	GLM	Sustained in part	R1 conditioned to equilibrium; excited-state case merged with U3 (covered at leading CFT order by DeepSeek's derivation; generic case open)
R3 audit broken: A4's T_∞ is entropic-clock-referred	GLM	Sustained in part	A4 compliance now conditional on the equilibrium/horizon identification (thermal time = modular time); calibration to be anchored at horizons
Erker clock-dissipation forces a label coupling	GLM (attack posed by the note)	**Overruled**	R4 survives: dissipation is per-tick (state property), stress-energy is a state functional; matter gravitates, label does not
Run 4 a strawman?	GLM	Overruled (fair), with a sharper control proposed	Run 5 added
Run 5 (lapse-as-state-functional coupling) diverges — apparent counterexample to the refined rule	Toy result	**Resolved as numerics**	Coupling-strength scan + step-halving: divergence vanishes at moderate coupling and shrinks with refinement (stiffness at the stasis coordinate singularity); run 4's real leak is step-independent by contrast
δ_lock correction covariance (parallel derivation)	DeepSeek	Delivered; verdict framing corrected by adjudication	Leading correction is local, covariant, bilinear (FHHPRV 2017); no foliation dependence at leading order; measure obstruction (i) remains in non-CFT settings. DeepSeek's "unimodular only at linear order" framing overruled: its own equations keep Λ as an integration constant at all orders — the trace enters via Bianchi + conservation exactly as

Attack	Source	Verdict	Disposition
			the unimodular⇄GR equivalence always works; what is new is the covariant bilinear completion of the source, not a change in what the entropic input determines
HT vs WTG selection (round-4 attack 3)	GLM	Answered	A2/A3 do not select HT — both formulations admit the operational clock; conventional classically, HT distinct only quantum-mechanically (canonical gravitational time), which ETRG-0 does not currently use. U4 stays a research direction; the named stronger question (does loop self-consistency require an unfrozen gravitational sector?) remains open

References

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